

Arrays counted up to renaming: the empirical recurrence for OEIS A205424

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Abstract

OEIS A205424 carries an empirical recurrence of order 5 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry imposes two conditions of different kinds: a condition on every subblock, which is local, and “new values introduced in row major order”, which is not — it makes the count a count of colourings up to renaming. The first is handled by a window of consecutive rows; the second by one extra integer, the number of values introduced so far, which is an exact summary of everything the whole prefix contributes. With both in the state the arrays are exactly the walks in a digraph on $S = 9$ vertices, so $a(n)$ is C -finite and whether the conjectured recurrence annihilates it is settled by a finite exact computation.

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1 The conjecture

OEIS A205424 is “Number of $(n+1) \times 2$ 0..3 arrays with the number of equal 2×2 subblock diagonal pairs and equal antidiagonal pairs differing from each horizontal or vertical neighbor, and new values 0..3 introduced in row major order.”. It has offset 1 and begins

15, 102, 870, 7566, 68342, 609086, 5514982, 49362414, ...

Because of the relabelling condition each array is counted once for the whole class of arrays that differ from it only by renaming the values, so these are colourings counted up to permutation of the colours. The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = a(n-1) + 73*a(n-2) + 39*a(n-3) - 384*a(n-4) - 432*a(n-5)$ for $n > 6$.

As of the “Last modified” line on the live entry (Mar 03 2018 13:57:19, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays of 2 columns over the alphabet $\{0, 1, 2, 3\}$, and asks two different kinds of thing of them at once.

The subblock condition is local. It says that every 2×2 subblock scores the number of its equal diagonal pairs plus the number of its equal antidiagonal pairs — a number in $\{0, 1, 2\}$ — and no subblock may score what the subblock beside it or the subblock above it scores. A 2×2 subblock spans 2 consecutive rows, so a window of 2 consecutive rows decides every subblock it closes, and nothing outside that window is consulted.

What is compared *across* windows is carried too: the state holds the scores of the whole last band, one number per subblock, since the band arriving next is compared with it subblock by subblock. With 1 subblock to a band that is a bounded amount of bookkeeping and no more.

The relabelling condition is not local. “New values introduced in row major order” says that, reading the array one row at a time and each row left to right, a value may appear for the first time only once every smaller value already has. Whether a given entry is allowed here therefore depends on the whole prefix of the array, not on any bounded window of it, and no amount of widening the window captures it. What saves it is that the prefix matters only through one number: let m be how many values have been introduced so far. A row $r_1 r_2 \cdots r_2$ may follow precisely when each $r_i \leq m$ at the moment it is read and each $r_i = m$ raises m by one. So m , an integer in $\{0, 1, \dots, 4\}$, is an exact summary of everything the condition needs, and it joins the state.

The vertices of the digraph are therefore the triples (the last 1 row, the count m , what the last band scored), an edge $u \rightarrow v$ meaning that v ’s new row may follow u . Of the 16 possible rows only those admissible from the current m appear, and states with identical futures are merged, which leaves $S = 9$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) = \iota^\top M^n \tau,$$

the exponent fixed by the entry’s own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+5} - \sum_i c_i M^{j+5-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = a(n-1) + 73a(n-2) + 39a(n-3) - 384a(n-4) - 432a(n-5)$$

for every $n > 6$.

Proof. The vector $w = q(M)\tau$ was formed by 5 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 6 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The bound is exact: at $n = 6$ the recurrence fails on the entry’s own published terms, so no larger range holds. $\square$

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry’s English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates the arrays one by one from the entry's own words — generating every array over the alphabet, discarding those that fail the relabelling condition, then testing each subblock directly — and compares the count with the published terms. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
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